

SOLIDWORKS API (.NET, VBA, C#) — Reference

(29/337): IBody2 (part 4 of 5)

SOLIDWORKS API Documentation (converted from CHM)

IBody2 (part 4 of 5) (Type)

IDisplay Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IDisplay Method (IBody2)

Obsolete. Superseded by [IBody2::Display3](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub IDisplay( _  
    ByVal Part As PartDoc, _  
    ByVal Color As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim Part As PartDoc  
Dim Color As System.Integer  
  
instance.IDisplay(Part, Color)
```

C#

```
void IDisplay(  
    PartDoc Part,  
    System.int Color  
)
```

C++/CLI

```
void IDisplay(  
& PartDoc^ Part,  
& System.int Color  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Part

Color

Visual Basic for Applications (VBA) Syntax

See [Body2::IDisplay](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

IDisplayWireFrameXOR Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IDisplayWireFrameXOR Method (IBody2)

Displays a temporary body in the given part's context in XOR mode.

.NET Syntax

Visual Basic (Declaration)

```
Sub IDisplayWireFrameXOR( _
    ByVal Part As PartDoc, _
    ByVal Color As System.Integer _
)
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim Part As PartDoc
Dim Color As System.Integer

instance.IDisplayWireFrameXOR(Part, Color)
```

C#

```
void IDisplayWireFrameXOR(
    PartDoc Part,
    System.int Color
)
```

C++/CLI

```
void IDisplayWireFrameXOR(
    & PartDoc^ Part,
    & System.int Color
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Part
Pointer to the [part](#)
Color
Color of part

Visual Basic for Applications (VBA) Syntax

See [Body2::IDisplayWireFrameXOR](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::DisplayWireFrameXOR Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IDraftBody Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IDraftBody Method (IBody2)

Obsolete. Superseded by [IBody2::IDraftBody2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IDraftBody( _
    ByVal NumOfFaces As System.Integer, _
    ByRef FaceList As Face2, _
    ByRef EdgeList As Edge, _
    ByVal DraftAngle As System.Double, _
    ByRef Dir As System.Double _
) As System.Boolean
```

```

Dim instance As IBody2
Dim NumOfFaces As System.Integer
Dim FaceList As Face2
Dim EdgeList As Edge
Dim DraftAngle As System.Double
Dim Dir As System.Double
Dim value As System.Boolean

value = instance.IDraftBody(NumOfFaces, FaceList, EdgeList, DraftAngle, Dir)

```

C#

```

System.bool IDraftBody(
    System.int NumOfFaces,
    ref Face2 FaceList,
    ref Edge EdgeList,
    System.double DraftAngle,
    ref System.double Dir
)

```

C++/CLI

```

System.bool IDraftBody(
    & System.int NumOfFaces,
    & Face2^% FaceList,
    & Edge^% EdgeList,
    & System.double DraftAngle,
    & System.double% Dir
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumOfFaces

FaceList

EdgeList

DraftAngle

Dir

Visual Basic for Applications (VBA) Syntax

See [Body2.IDraftBody](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

SOLIDWORKS API Help

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IDraftBody2 Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IDraftBody2 Method (IBody2)

Adds drafts to the specified faces on a temporary body. This method modifies the body.

.NET Syntax

Visual Basic (Declaration)

```

Function IDraftBody2( _
    ByVal NumOfFaces As System.Integer, _
    ByRef FaceList As Face2, _
    ByRef EdgeList As Edge, _
    ByVal BasePoint As MathPoint, _

```

```

        ByVal DraftAngle As System.Double, _
        ByRef Dir As System.Double _
    ) As System.Boolean

```

Visual Basic (Usage)



```

Dim instance As IBody2
Dim NumOfFaces As System.Integer
Dim FaceList As Face2
Dim EdgeList As Edge
Dim BasePoint As MathPoint
Dim DraftAngle As System.Double
Dim Dir As System.Double
Dim value As System.Boolean

value = instance.IDraftBody2(NumOfFaces, FaceList, EdgeList, BasePoint, DraftAngle, Dir)

```

C#

```

System.bool IDraftBody2(
    System.int NumOfFaces,
    ref Face2 FaceList,
    ref Edge EdgeList,
    MathPoint BasePoint,
    System.double DraftAngle,
    ref System.double Dir
)

```

C++/CLI

```

System.bool IDraftBody2(
    & System.int NumOfFaces,
    & Face2^% FaceList,
    & Edge^% EdgeList,
    & MathPoint^ BasePoint,
    & System.double DraftAngle,
    & System.double% Dir
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumOfFaces
Number of faces to draft

FaceList
Array of [faces](#) to draft

EdgeList
Array of [edges](#), one for each face, along which to apply the drafts

BasePoint
Pointer to a [MathPoint](#) object (x,y,z values of the base point)

DraftAngle
Draft angle

Dir
Array of 3 doubles (x, y, z), a vector which specifies the direction of the draft

Return Value

True if drafts are applied, false if not

Visual Basic for Applications (VBA) Syntax

See [Body2::IDraftBody2](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::DraftBody2 Method](#)

Availability

IExtendSurface Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IExtendSurface Method (IBody2)

Creates a new temporary body by extending the selected edges.

.NET Syntax

Visual Basic (Declaration)

```
Function IExtendSurface( _
    ByVal EdgeCount As System.Integer, _
    ByRef EdgesToExtend As Edge, _
    ByVal ExtendLinear As System.Boolean, _
    ByVal EndCondition As System.Integer, _
    ByVal Dist As System.Double, _
    ByVal PUpToVtx As Vertex, _
    ByVal PUpToFace As Face _
) As Body2
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim EdgeCount As System.Integer
Dim EdgesToExtend As Edge
Dim ExtendLinear As System.Boolean
Dim EndCondition As System.Integer
Dim Dist As System.Double
Dim PUpToVtx As Vertex
Dim PUpToFace As Face
Dim value As Body2
```

```
value = instance.IExtendSurface(EdgeCount, EdgesToExtend, ExtendLinear, EndCondition, Dist,
PUpToVtx, PUpToFace)
```

C#

```
Body2 IExtendSurface(
    System.int EdgeCount,
    ref Edge EdgesToExtend,
    System.bool ExtendLinear,
    System.int EndCondition,
    System.double Dist,
    Vertex PUpToVtx,
    Face PUpToFace
)
```

C++/CLI

```
Body2^ IExtendSurface(
    & System.int EdgeCount,
    & Edge^% EdgesToExtend,
    & System.bool ExtendLinear,
    & System.int EndCondition,
    & System.double Dist,
    & Vertex^ PUpToVtx,
    & Face^ PUpToFace
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

EdgeCount

Number of edges to extend

EdgesToExtend

Array of the selected [edges](#) to extend

ExtendLinear

True to extend the selected edges linearly along the tangent of the face at the edges; false to extend the selected edges in the same direction as the face

EndCondition

- 0 = Extend selected edges per value specified for Dist
- 1 = Extend selected edges to PUpToVtx

- 2 = Extend selected edges to PpUpToFace

Dist

Distance by which to extend the selected edges (see **Remarks**)

PUpToVtx

[Vertex](#) up to which to extend the selected edges (see **Remarks**)

PUpToFace

[Face](#) up to which to extend the selected edges (see **Remarks**)

Return Value

Pointer to the newly created [IBody2](#) object

Visual Basic for Applications (VBA) Syntax

See [Body2::IExtendSurface](#).

Remarks

This method supports surface bodies only.

You can extend the edges by a specified distance or up to a vertex or up to a face.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::ExtendSurface Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14

SOLIDWORKS API Help

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IGetBodyBox Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IGetBodyBox Method (IBody2)

Gets the bounding box for this body.

.NET Syntax

Visual Basic (Declaration)

```
Sub IGetBodyBox( _
    ByRef BoxCorners As System.Double _
)
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim BoxCorners As System.Double

instance.IGetBodyBox(BoxCorners)
```

C#

```
void IGetBodyBox(
    ref System.double BoxCorners
)
```

C++/CLI

```
void IGetBodyBox(
    & System.double% BoxCorners
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BoxCorners

Pointer to an array of 6 doubles representing the extents of the bounding box (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Body2::IGetBodyBox](#).

Remarks

The X,Y,Z points returned by SOLIDWORKS are the lower and upper diagonal corners which bound the body with the box sides parallel to the X Y and Z axes. The box dimensions returned by SOLIDWORKS enclose the body and are typically close to the minimum possible size (this is typical, but not always true).

The return value is an array of doubles as follows:

[*XCorner1*, *YCorner1*, *ZCorner1*, *XCorner2*, *YCorner2*, *ZCorner2*]

IMPORTANT: The values returned are approximate and should not be used for comparison or calculation purposes. Furthermore, the bounding box may vary after rebuilding the model.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetBodyBox Method](#)
[IModelDocExtension::SetVisibleBox Method](#)
[IModelDocExtension::RemoveVisibleBox Method](#)
[IModelDocExtension::GetVisibleBox Method](#)
[IFeature::GetBox Method](#)
[IFeature::IGetBox Method](#)
[IFace2::GetBox Method](#)
[IFace2::IGetBox Method](#)
[IComponent2::GetBox Method](#)
[IComponent2::IGetBox Method](#)
[IAssemblyDoc::GetBox Method](#)
[IAssemblyDoc::IGetBox Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IGetEdges Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IGetEdges Method (IBody2)

Gets the edges for this body.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetEdges( _  
    ByVal Count As System.Integer _  
) As Edge
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim Count As System.Integer  
Dim value As Edge  
  
value = instance.IGetEdges(Count)
```

C#

```
Edge IGetEdges(  
    System.int Count  
)
```

```
Edge^ IGetEdges(
& System.int Count
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count
Number of edges

Return Value

- in-process, unmanaged C++: Pointer to an array of [edges](#) of size Count
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [IBody2::GetEdgeCount](#) to determine the size of the array.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetEdges Method](#)
[IModelDocExtension::SelectAll Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IGetExcessBodyArray Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IGetExcessBodyArray Method (IBody2)

Gets the excess bodies after sewing.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetExcessBodyArray() As Body2
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim value As Body2
```

```
value = instance.IGetExcessBodyArray()
```

C#

```
Body2 IGetExcessBodyArray()
```

C++/CLI

```
Body2^ IGetExcessBodyArray();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of excess [bodies](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::IGetExcessBodyCount Method](#)
[IBody2::GetExcessBodyArray Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetExcessBodyCount Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IGetExcessBodyCount Method (IBody2)

Gets the number of excess bodies.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetExcessBodyCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As System.Integer  
  
value = instance.IGetExcessBodyCount()
```

C#

```
System.int IGetExcessBodyCount()
```

C++/CLI

```
System.int IGetExcessBodyCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of excess bodies

Visual Basic for Applications (VBA) Syntax

See [Body2::IGetExcessBodyCount](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetExcessBodyArray Method](#)
[IBody2::IGetExcessBodyArray Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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IGetFaces Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IGetFaces Method (IBody2)

Gets all of the faces on the body.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFaces( _  
    ByVal Count As System.Integer _  
    ) As Face2
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim Count As System.Integer  
Dim value As Face2  
  
value = instance.IGetFaces(Count)
```

C#

```
Face2 IGetFaces(  
    System.int Count  
)
```

C++/CLI

```
Face2^ IGetFaces(  
&    System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of faces on the body (see **Remarks**)

Return Value

- in-process, unmanaged C++: Pointer to an array of [faces](#) on the body
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

Call [IBody2::GetFaceCount](#) to get the value of Count.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetFaces Method](#)
[IBody2::IGetFirstFace Method](#)
[IBody2::GetFirstFace Method](#)

Availability

SOLIDWORKS 2008 SP1, Revision Number 16.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetFeatures Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IGetFeatures Method (IBody2)

Gets the features in this body in a multibody sheet metal part.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFeatures( _  
    ByVal NumberOfFeatures As System.Integer _  
    ) As Feature
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim NumberOfFeatures As System.Integer
Dim value As Feature

value = instance.IGetFeatures(NumberOfFeatures)
```

C#

```
Feature IGetFeatures(
    System.int NumberOfFeatures
)
```

C++/CLI

```
Feature^ IGetFeatures(
    & System.int NumberOfFeatures
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumberOfFeatures
Number of features in this body in a multibody sheet metal part

Return Value

- in-process, unmanaged C++: Pointer to an array of [features](#) in this body in a multibody sheet metal part
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [IBody2::GetFeatureCount](#) to get NumberOfFeatures.

The features of a body in a multibody sheet metal part are located in the solid bodies folder in the FeatureManager design tree.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetFeatures Method](#)
[IBodyFolder Interface](#)
[IBodyFolder::GetBodyCount Method](#)
[IBodyFolder::GetBodies Method](#)
[IBodyFolder::IGetBodies Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

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IGetFirstFace Method (IBody2)

See Also [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IGetFirstFace Method (IBody2)

Finds the first face in a body and returns the face.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstFace() As Face2
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim value As Face2
```

```
value = instance.IGetFirstFace()
```

C#

```
Face2 IGetFirstFace()
```

C++/CLI

```
Face2^ IGetFirstFace();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

First [face](#) in the body

Visual Basic for Applications (VBA) Syntax

See [Body2::IGetFirstFace](#).

Example

[Traverse Bodies \(C++\)](#)

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetFaceCount Method](#)
[IBody2::GetFirstFace Method](#)
[IBody2::EnumFaces Method](#)
[IBody2::!DeleteBlends2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IGetFirstSelectedFace Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IGetFirstSelectedFace Method (IBody2)

Gets the first selected face on this body. For use with [IBody2::IGetProcessedBodyWithSelfFace](#) and intended for IGES routines.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstSelectedFace() As Face2
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As Face2
```

```
value = instance.IGetFirstSelectedFace()
```

C#

```
Face2 IGetFirstSelectedFace()
```

C++/CLI

```
Face2^ IGetFirstSelectedFace();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

First selected [face](#)

Visual Basic for Applications (VBA) Syntax

See [Body2::IGetFirstSelectedFace](#).

Example

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

Remarks

Do not use this method for general selection handling. If you want to handle items selected by the user or items selected with [IModelDocExtension::SelectByID2](#), use [ISelectionMgr::GetSelectedObject6](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::IGetFirstSelectedFace Method](#)
[IBody2::IGetNextSelectedFace Method](#)
[IBody2::IGetSelectedFaceCount Method](#)
[IBody2::IGetSelectedFaces Method](#)
[IBody2::IGetSelectedFaces Method](#)
[IBody2::IGetProcessedBodyWithSelfFace Method](#)
[IBody2::IGetProcessedBodyWithSelfFace Method](#)

SOLIDWORKS API Help

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IGetIntersectionEdgeCount Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IGetIntersectionEdgeCount Method (IBody2)

Gets the number of intersecting edges between this body and the specified tool body.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetIntersectionEdgeCount( _  
    ByVal ToolBodyIn As Body2 _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim ToolBodyIn As Body2  
Dim value As System.Integer  
  
value = instance.IGetIntersectionEdgeCount(ToolBodyIn)
```

C#

```
System.int IGetIntersectionEdgeCount(  
    Body2 ToolBodyIn  
)
```

C++/CLI

```
System.int IGetIntersectionEdgeCount(  
    & Body2^ ToolBodyIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ToolBodyIn

Pointer to the temporary body used to perform the intersection

Return Value

Number of intersection edges

Visual Basic for Applications (VBA) Syntax

See [Body2::IGetIntersectionEdgeCount](#).

Remarks

Use the return value from this method to determine the size of the array returned by [IBody2::IGetIntersectionEdges](#).

If the two bodies are in an assembly, then `IBody2::GetIntersectionEdges` and `IBody2::IGetIntersectionEdges` generate a list of the intersection edges between the two bodies. To do this, the second body must be transformed in its coordinate space so that it is positioned the same with respect to the first body as it is in assembly space.

To align the two bodies before calling `IBody2::GetIntersectionEdges`, `IBody2::IGetIntersectionEdges`, or `Body2::IGetIntersectionEdgeCount`, calculate the transformation from the first body to the second body and apply the transform to the second body using [IBody2::ApplyTransform](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

SOLIDWORKS API Help

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IGetIntersectionEdges Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IGetIntersectionEdges Method (IBody2)

Obsolete. Superseded by [IBody2::GetIntersectionEdges2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetIntersectionEdges( _  
    ByVal ToolBodyIn As Body2 _  
) As Edge
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim ToolBodyIn As Body2  
Dim value As Edge
```

```
value = instance.IGetIntersectionEdges(ToolBodyIn)
```

C#

```
Edge IGetIntersectionEdges(  
    Body2 ToolBodyIn  
)
```

C++/CLI

```
Edge^ IGetIntersectionEdges(  
& Body2^ ToolBodyIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ToolBodyIn

Temporary [body](#) used to perform the intersection

Return Value

- in-process, unmanaged C++: Pointer to an array of [edges](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported
See [In-process Methods](#) for details about this type of method.

Remarks

This method imprints a set of edges on both of these temporary bodies and returns the edges in an unordered list as shown below. The total number of edges in this array is twice the value returned from [IBody2::IGetIntersectionEdgeCount](#).

[*Edge1imprintedOnTarget*, *Edge1imprintedOnTool*, *Edge2imprintedOnTarget*, *Edge2imprintedOnTool*]

where the target body is the IBody2 object used to call this method and the tool body is passed into this method as the first argument.

If the two bodies are in an assembly, then IBody2::GetIntersectionEdges generates a list of the intersection edges between the two bodies. To do this, the second body must be transformed in its coordinate space so that it is positioned the same with respect to the first body as it is in assembly space.

To align the two bodies before calling IBody2::GetIntersectionEdges or IBody2::GetIntersectionEdgeCount, calculate the transformation from the first body to the second body and set this as the transform for the second body using [IBody2::ApplyTransform](#).

In the case of a tangency condition (for example, a planar face contacting the cylindrical face of a cylinder), this method returns a single edge along the tangency.

You might also find that [IBody2::!Operations2](#) provides an adequate solution. This method intersects a sheet body with a target body.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetIntersectionEdges Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IGetMassProperties Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IGetMassProperties Method (IBody2)

Gets the mass properties of this body.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetMassProperties( _  
    ByVal Density As System.Double _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim Density As System.Double  
Dim value As System.Double  
  
value = instance.IGetMassProperties(Density)
```

C#

```
System.double IGetMassProperties(  
    System.double Density  
)
```

C++/CLI

```
System.double IGetMassProperties(  
& System.double Density  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Density

Density to use for the mass property calculations on this body

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

This method is intended for obtaining the mass properties of temporary objects but may also be used with the SOLIDWORKS [IBody2](#) object that is created by the user. To get the mass properties of the SOLIDWORKS IBody2 object created by the user, you can also use [IModelDocExtension::IGetMassProperties](#), which uses the density currently set for the body's material.

The return value is an array of doubles as follows:

Solid body

```
[ CenterOfMassX, CenterOfMassY, CenterOfMassZ, Volume, Area, Mass(Volume*density), MomXX, MomYY, MomZZ, MomXY, MomZX, MomYZ ]
```

Sheet body

```
[ CenterOfMassX, CenterOfMassY, CenterOfMassZ, Area, Circumference, Mass(Area*density), MomXX, MomYY, MomZZ, MomXY, MomZX, MomYZ ]
```

Wire body

```
[ CenterOfMassX, CenterOfMassY, CenterOfMassZ, Length, 0, Mass(Length*density), MomXX, MomYY, MomZZ, MomXY, MomZX, MomYZ ]
```

You can use [ISldWorks::GetUserPreferenceDoubleValue](#) to get the density used by the SOLIDWORKS part. Consistent with all other SOLIDWORKS API functions, units are metric unless otherwise specified.

SOLIDWORKS returns information (such as the center of mass) in relation to where the body was created. For example, if you create a block in a part file that is centered at (0,0,0), then SOLIDWORKS returns the center of mass as (0,0,0). If you then use this part at some random location within an assembly, and the body is obtained from the assembly component object using [IComponent2::IGetBody](#), then SOLIDWORKS still returns the center of mass as (0,0,0). If you need to determine the body's center of mass in relation to the assembly coordinate system, you need to multiply the component transform and the center of mass coordinates (see [IComponent2::Transform2](#)).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetMassProperties Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetMaterialPropertyValuesForFace Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IGetMaterialPropertyValuesForFace Method (IBody2)

Gets the color of the specified face.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetMaterialPropertyValuesForFace( _  
    ByVal FaceIn As System.Object _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim FaceIn As System.Object  
Dim value As System.Double  
  
value = instance.IGetMaterialPropertyValuesForFace(FaceIn)
```

C#

```
System.double IGetMaterialPropertyValuesForFace(  
    System.object FaceIn  
)
```

C++/CLI

```
System.double IGetMaterialPropertyValuesForFace(  
& System.Object^ FaceIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FaceIn

Face from which to get its color

Return Value

Color of the face

Visual Basic for Applications (VBA) Syntax

See [Body2::IGetMaterialPropertyValuesForFace](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IComponent2::IGetMaterialPropertyValuesForFace Method](#)
[IBody2::GetMaterialIdName2 Method](#)
[IBody2::GetMaterialPropertyName2 Method](#)
[IBody2::GetMaterialUserName2 Method](#)
[IBody2::IRemoveMaterialProperty Method](#)
[IBody2::RemoveMaterialProperty Method](#)
[IBody2::SetMaterialIdName2 Method](#)
[IBody2::SetMaterialProperty Method](#)
[IBody2::SetMaterialUserName2 Method](#)
[IBody2::IMaterialPropertyValues2 Property](#)
[IBody2::MaterialPropertyValues2 Property](#)

Availability

SOLIDWORKS 2003 SP1, Revision Number 11.1

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetMiddleSurface Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IGetMiddleSurface Method (IBody2)

Inserts a midsurface in a body.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetMiddleSurface( _  
    ByVal PlacementPercentage As System.Double, _  
    ByRef Face1List As System.Object, _  
    ByRef Face2List As System.Object, _  
    ByRef Thickness As System.Object, _  
    ByRef MiddleSurfaceBody As Body2 _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim PlacementPercentage As System.Double  
Dim Face1List As System.Object  
Dim Face2List As System.Object  
Dim Thickness As System.Object  
Dim MiddleSurfaceBody As Body2  
Dim value As System.Integer
```

```
value = instance.IGetMiddleSurface(PlacementPercentage, Face1List, Face2List, Thickness,  
MiddleSurfaceBody)
```

C#

```
System.int IGetMiddleSurface(  
    System.double PlacementPercentage,  
    out System.object Face1List,  
    out System.object Face2List,  
    out System.object Thickness,
```

```
out Body2 MiddleSurfaceBody
)
```

C++/CLI

```
System.int IGetMiddleSurface(
& System.double PlacementPercentage,
& [Out] System.Object^ Face1List,
& [Out] System.Object^ Face2List,
& [Out] System.Object^ Thickness,
& [Out] Body2^ MiddleSurfaceBody
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PlacementPercentage

Position where to insert the midsurface in the body; value of 50 indicates to insert the surface in the middle

Face1List

Array of faces into which the midsurface will be or was inserted; you can provide an array of faces for this parameter and Face2List; if you do not, then the faces are determined internally

Face2List

Array of faces into which the midsurface will be or was inserted; you can provide an array of faces for this parameter and Face1List; if you do not, then the faces are determined internally

Thickness

Array containing minimum and maximum thickness of sheet metal for pairs of faces

[MiddleSurfaceBody](#)

[Middle surface body](#)

Return Value

0 if no errors; -1 if errors

Visual Basic for Applications (VBA) Syntax

See [Body2::IGetMiddleSurface](#).

See Also

[IBody2 Interface](#)

[IBody2 Members](#)

[IBody2::GetMiddleSurface Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetNextSelectedFace Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IGetNextSelectedFace Method (IBody2)

Gets the next selected face. For use with [IBody2::GetProcessedBodyWithSelfFace](#) and intended for IGES routines.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetNextSelectedFace() As Face2
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim value As Face2
```

```
value = instance.IGetNextSelectedFace()
```

C#

```
Face2 IGetNextSelectedFace()
```

C++/CLI

```
Face2^ IGetNextSelectedFace();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the selected [face](#)

Visual Basic for Applications (VBA) Syntax

See [Body2::IGetNextSelectedFace](#).

Remarks

Do not use this method for general selection handling. If you want to handle items selected by the user or items selected with [IModelDocExtension::SelectById2](#), use [ISelectionMgr::GetSelectedObject6](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::IGetFirstSelectedFace Method](#)
[IBody2::IGetNextSelectedFace Method](#)
[IBody2::IGetSelectedFaceCount Method](#)
[IBody2::IGetSelectedFaces Method](#)
[IBody2::IGetSelectedFaces Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetProcessedBody Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IGetProcessedBody Method (IBody2)

Obsolete. Superseded by [IBody2::GetProcessedBody2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetProcessedBody() As Body2
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As Body2
```

```
value = instance.IGetProcessedBody()
```

C#

```
Body2 IGetProcessedBody()
```

C++/CLI

```
Body2^ IGetProcessedBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Body2::IGetProcessedBody](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetProcessedBodyWithSelfFace Method (IBody2)

[See Also](#)



Gets a processed body.

 .NET Syntax

Visual Basic (Declaration)

```
Function IGetProcessedBodyWithSelFace() As Body2
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim value As Body2

value = instance.IGetProcessedBodyWithSelFace()
```

C#

```
Body2 IGetProcessedBodyWithSelFace()
```

C++/CLI

```
Body2^ IGetProcessedBodyWithSelFace();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [IBody2](#) object; this body is a copy of the body for this part

 Visual Basic for Applications (VBA) Syntax

See [Body2::IGetProcessedBodyWithSelFace](#).

 Remarks

This method requires that you preselect a face.

 See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetProcessedBody2 Method](#)
[IBody2::GetProcessedBodyWithSelFace Method](#)
[IGetSelectedFaces Method](#)
[IBody2::IGetNextSelectedFace Method](#)
[IBody2::IGetFirstFace Method](#)
[IBody2::GetSelectedFaces Method](#)
[IBody2::GetNextSelectedFace Method](#)
[IBody2::GetFirstSelectedFace Method](#)
[IBody2::GetSelectedFaceCount Method](#)

 Availability

SOLIDWORKS 2001Plus FCS Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetSelectedFaces Method (IBody2)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > IBody2 Interface : IGetSelectedFaces Method (IBody2)

Gets the selected faces. For use with [IBody2::GetProcessedBodyWithSelFace](#) and intended for IGES routines.

 .NET Syntax

Visual Basic (Declaration)

```
Function IGetSelectedFaces( _
    ByVal SelectedFaceCount As System.Integer _
) As Face2
```

```
Dim instance As IBody2
Dim SelectedFaceCount As System.Integer
Dim value As Face2

value = instance.IGetSelectedFaces(SelectedFaceCount)
```

C#

```
Face2 IGetSelectedFaces(
    System.int SelectedFaceCount
)
```

C++/CLI

```
Face2^ IGetSelectedFaces(
    & System.int SelectedFaceCount
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SelectedFaceCount
Number of selected faces

Return Value

- in-process, unmanaged C++: Pointer to an array of selected [faces](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported
See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [IBody2::GetSelectedFaces](#) to get SelectedFaceCount.

Do not use this method for general selection handling. If you want to handle items selected by the user or items selected with [IModelDocExtension::SelectById2](#), use [ISelectionMgr::GetSelectedObject6](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetSelectedFaces Method](#)
[IBody2::GetSelectedFaceCount Method](#)
[IBody2::GetFirstSelectedFace Method](#)
[IBody2::GetNextSelectedFace Method](#)
[IBody2::IGetFirstSelectedFace Method](#)
[IBody2::IGetNextSelectedFace Method](#)
[IBody2::GetProcessedBodyWithSelfFace Method](#)
[IBody2::IGetProcessedBodyWithSelfFace Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetSheetBody Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IGetSheetBody Method (IBody2)

Gets a sheet (surface) body in this body.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSheetBody( _
    ByVal Index As System.Integer _
) As Body2
```

Visual Basic (Usage)

```
Dim instance As IBody2
Dim Index As System.Integer
Dim value As Body2

value = instance.IGetSheetBody(Index)
```

C#

```
Body2 IGetSheetBody(
    System.int Index
)
```

C++/CLI

```
Body2^ IGetSheetBody(
    & System.int Index
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Index of sheet body

Return Value

Pointer to the sheet [body](#) corresponding to the index

Visual Basic for Applications (VBA) Syntax

See [Body2::IGetSheetBody](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::IGetSheetBody Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetTessellation Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IGetTessellation Method (IBody2)

Gets the [ITessellation](#) object.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTessellation( _
    ByVal NumOfFaces As System.Integer, _
    ByRef FaceList As Face2 _
) As Tessellation
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim NumOfFaces As System.Integer
Dim FaceList As Face2
Dim value As Tessellation

value = instance.IGetTessellation(NumOfFaces, FaceList)
```

C#

```
Tessellation IGetTessellation(  
    System.int NumOfFaces,  
    ref Face2 FaceList  
)
```

C++/CLI

```
Tessellation^ IGetTessellation(  
& System.int NumOfFaces,  
& Face2^% FaceList  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumOfFaces

Number of faces to tessellate

FaceList

Array of faces to tessellate; if this is NULL, then SOLIDWORKS will tessellate the body

Return Value

Pointer to the [ITessellation](#) object

Visual Basic for Applications (VBA) Syntax

See [Body2::IGetTessellation](#).

Example

[Tessellate a Body \(C#\)](#)

See Also

[IBody2 Interface](#)

[IBody2 Members](#)

[IBody2::GetTessellation Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetTrackingIDs Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IGetTrackingIDs Method (IBody2)

Gets the [tracking IDs assigned to this body](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTrackingIDs( _  
    ByVal TrackingCookie As System.Integer, _  
    ByVal Count As System.Integer, _  
    ByRef TrackingIDs As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)

 Copy Code

```

Dim instance As IBody2
Dim TrackingCookie As System.Integer
Dim Count As System.Integer
Dim TrackingIDs As System.Integer
Dim value As System.Integer

value = instance.IGetTrackingIDs(TrackingCookie, Count, TrackingIDs)

```

C#

```

System.int IGetTrackingIDs(
    System.int TrackingCookie,
    System.int Count,
    out System.int TrackingIDs
)

```

C++/CLI

```

System.int IGetTrackingIDs(
    & System.int TrackingCookie,
    & System.int Count,
    [Out] System.int TrackingIDs
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TrackingCookie

Cookie obtained from [ISldWorks::RegisterTrackingDefinition](#)

Count

Number of tracking IDs

TrackingIDs

- in-process, unmanaged C++: Pointer to an array of tracking IDs on this body

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

Status as defined by [swTrackingIDError_e](#)

Remarks

Before calling this method, call [IBody2::GetTrackingIDsCount](#) to determine Count.

See [Tracking IDs](#) for more information.

See Also

[IBody2 Interface](#)

[IBody2 Members](#)

[IBody2::GetTrackingIDs Method](#)

[IBody2::RemoveTrackingID Method](#)

[IBody2::SetTrackingID Method](#)

[IModelDocExtension::FindTrackedObjects Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetVertices Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IGetVertices Method (IBody2)

Gets the vertices in this body.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetVertices( _  
    ByVal Count As System.Integer _  
    ) As Vertex
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim Count As System.Integer  
Dim value As Vertex  
  
value = instance.IGetVertices(Count)
```

C#

```
Vertex IGetVertices(  
    System.int Count  
)
```

C++/CLI

```
Vertex^ IGetVertices(  
&    System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count
Number of vertices in this body

Return Value

- in-process, unmanaged C++: Pointer to an array of [vertices](#) in this body
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [IBody2::GetVertexCount](#) to determine the size of the array.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[GetVertices Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IHide Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IHide Method (IBody2)

Hides a temporary body using the specified part's context.

.NET Syntax

Visual Basic (Declaration)

```
Sub IHide( _  
    ByVal Part As PartDoc _  
    )
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim Part As PartDoc  
  
instance.IHide(Part)
```

C#

```
void IHide(  
    PartDoc Part  
)
```

C++/CLI

```
void IHide(  
& PartDoc^ Part  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Part
[IPartDoc](#)

Visual Basic for Applications (VBA) Syntax

See [Body2::IHide](#).

Remarks

While SOLIDWORKS is displaying your IBody2 object using [IBody2::Display3](#), you cannot release it explicitly or implicitly. Before releasing or allowing your IBody2 object to be released, you must call this method to prevent it from being displayed.

COM applications should avoid explicitly releasing the IBody2 object by calling IBody2->Release() while it is being displayed by SOLIDWORKS.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::Hide Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IMatchedBoolean Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IMatchedBoolean Method (IBody2)

Obsolete. Superseded by [IBody2::IMatchedBoolean3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IMatchedBoolean(  
    ByVal OperationType As System.Integer, _  
    ByVal ToolBody As Body2, _  
    ByVal NumOfMatchingFaces As System.Integer, _  
    ByRef FaceList1 As Face2, _  
    ByRef FaceList2 As Face2, _  
    ByRef ErrorCode As System.Integer _  
) As EnumBodies2
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim OperationType As System.Integer  
Dim ToolBody As Body2  
Dim NumOfMatchingFaces As System.Integer  
Dim FaceList1 As Face2  
Dim FaceList2 As Face2  
Dim ErrorCode As System.Integer
```

```
Dim value As EnumBodies2
```

```
value = instance.IMatchedBoolean(OperationType, ToolBody, NumOfMatchingFaces, FaceList1,  
FaceList2, ErrorCode)
```

C#

```
EnumBodies2 IMatchedBoolean(  
    System.int OperationType,  
    Body2 ToolBody,  
    System.int NumOfMatchingFaces,  
    ref Face2 FaceList1,  
    ref Face2 FaceList2,  
    out System.int ErrorCode  
)
```

C++/CLI

```
EnumBodies2^ IMatchedBoolean(  
& System.int OperationType,  
& Body2^ ToolBody,  
& System.int NumOfMatchingFaces,  
& Face2^% FaceList1,  
& Face2^% FaceList2,  
& [Out] System.int ErrorCode  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OperationType

ToolBody

NumOfMatchingFaces

FaceList1

FaceList2

ErrorCode

Visual Basic for Applications (VBA) Syntax

See [Body2::IMatchedBoolean](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IMatchedBoolean2 Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IMatchedBoolean2 Method (IBody2)

Obsolete. Superseded by [IBody2::IMatchedBoolean3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IMatchedBoolean2( _  
    ByVal OperationType As System.Integer, _  
    ByVal ToolBodyCount As System.Integer, _  
    ByRef ToolBodyArr As Body2, _  
    ByVal NumOfMatchingFaces As System.Integer, _  
    ByRef FaceList1 As Face2, _  
    ByRef FaceList2 As Face2, _  
    ByRef ErrorCode As System.Integer _  
) As EnumBodies2
```

```

Dim instance As IBody2
Dim OperationType As System.Integer
Dim ToolBodyCount As System.Integer
Dim ToolBodyArr As Body2
Dim NumOfMatchingFaces As System.Integer
Dim FaceList1 As Face2
Dim FaceList2 As Face2
Dim ErrorCode As System.Integer
Dim value As EnumBodies2

value = instance.IMatchedBoolean2(OperationType, ToolBodyCount, ToolBodyArr, NumOfMatchingFaces,
FaceList1, FaceList2, ErrorCode)

```

C#

```

EnumBodies2 IMatchedBoolean2(
    System.int OperationType,
    System.int ToolBodyCount,
    ref Body2 ToolBodyArr,
    System.int NumOfMatchingFaces,
    ref Face2 FaceList1,
    ref Face2 FaceList2,
    out System.int ErrorCode
)

```

C++/CLI

```

EnumBodies2^ IMatchedBoolean2(
    & System.int OperationType,
    & System.int ToolBodyCount,
    & Body2^% ToolBodyArr,
    & System.int NumOfMatchingFaces,
    & Face2^% FaceList1,
    & Face2^% FaceList2,
    & [Out] System.int ErrorCode
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OperationType

ToolBodyCount

ToolBodyArr

NumOfMatchingFaces

FaceList1

FaceList2

ErrorCode

Visual Basic for Applications (VBA) Syntax

See [Body2::IMatchedBoolean2](#).

See Also

[IBody2 Interface](#)

[IBody2 Members](#)

IMatchedBoolean3 Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IMatchedBoolean3 Method (IBody2)

Obsolete. Superseded by [IBody2::IMatchedBoolean4](#).

Visual Basic (Declaration)

```
Function IMatchedBoolean3( _
    ByVal OperationType As System.Integer, _
    ByVal ToolBodyCount As System.Integer, _
    ByRef ToolBodyArr As Body2, _
    ByVal NumOfMatchingFaces As System.Integer, _
    ByRef FaceList1 As Face2, _
    ByRef FaceList2 As Face2, _
    ByRef ErrorCode As System.Integer _
) As EnumBodies2
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim OperationType As System.Integer
Dim ToolBodyCount As System.Integer
Dim ToolBodyArr As Body2
Dim NumOfMatchingFaces As System.Integer
Dim FaceList1 As Face2
Dim FaceList2 As Face2
Dim ErrorCode As System.Integer
Dim value As EnumBodies2
```

```
value = instance.IMatchedBoolean3(OperationType, ToolBodyCount, ToolBodyArr, NumOfMatchingFaces,
FaceList1, FaceList2, ErrorCode)
```

C#

```
EnumBodies2 IMatchedBoolean3(
    System.int OperationType,
    System.int ToolBodyCount,
    ref Body2 ToolBodyArr,
    System.int NumOfMatchingFaces,
    ref Face2 FaceList1,
    ref Face2 FaceList2,
    out System.int ErrorCode
)
```

C++/CLI

```
EnumBodies2^ IMatchedBoolean3(
    & System.int OperationType,
    & System.int ToolBodyCount,
    & Body2^% ToolBodyArr,
    & System.int NumOfMatchingFaces,
    & Face2^% FaceList1,
    & Face2^% FaceList2,
    & [Out] System.int ErrorCode
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OperationType

One of these operation types:

- SWBODYADD
- SWBODYCUT
- SWCODYINTERSECT

ToolBodyCount

Number of bodies

ToolBodyArr

Array of bodies of size ToolBodyCount

NumOfMatchingFaces

Number of matching faces

FaceList1

First face list (see **Remarks**)

FaceList2

Second face list (see **Remarks**)

ErrorCode

Error indicated as defined in [swBodyOperationError_e](#)

Return Value

Pointer to the [IEnumBodies2](#) object for a list of matches

Visual Basic for Applications (VBA) Syntax

See [Body2::IMatchedBoolean3](#).

Remarks

The concept of match means that the caller tells the boolean operator beforehand which faces can be considered to coincide. Basically, the caller performs part of the boolean operation.

Sometimes the application knows that two faces match because of the way the bodies are constructed; i.e., the application knows which faces are intended to match.

Having a list of matching face pairs may allow the matched boolean operator to resolve other geometric operations that otherwise it would not be able to work out. In general, providing matched faces speeds up the boolean operation and makes results more reliable.

The arguments FaceList1 and FaceList2 arguments can be empty lists. If matching face pairs are passed in, these faces must match such that:

- the surface geometry is coinciding.
- for each edge in a face, there is an edge in the other face that coincides

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::MatchedBoolean3 Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IMatchedBoolean4 Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IMatchedBoolean4 Method (IBody2)

Performs a matched boolean on the specified bodies and supports an optional list of faces that match exactly.

.NET Syntax

Visual Basic (Declaration)

```
Function IMatchedBoolean4( _  
    ByVal OperationType As System.Integer, _  
    ByVal ToolBodyCount As System.Integer, _  
    ByRef ToolBodyArr As Body2, _  
    ByVal NumOfMatchingFaces As System.Integer, _  
    ByRef FaceList1 As Face2, _  
    ByRef FaceList2 As Face2, _  
    ByVal MatchingTolerance As System.Double, _  
    ByRef ErrorCode As System.Integer _  
    ) As EnumBodies2
```

Visual Basic (Usage)

```
Dim instance As IBody2  
Dim OperationType As System.Integer  
Dim ToolBodyCount As System.Integer  
Dim ToolBodyArr As Body2
```



```

Dim NumOfMatchingFaces As System.Integer
Dim FaceList1 As Face2
Dim FaceList2 As Face2
Dim MatchingTolerance As System.Double
Dim ErrorCode As System.Integer
Dim value As EnumBodies2

```

```

value = instance.IMatchedBoolean4(OperationType, ToolBodyCount, ToolBodyArr, NumOfMatchingFaces,
FaceList1, FaceList2, MatchingTolerance, ErrorCode)

```

C#

```

EnumBodies2 IMatchedBoolean4(
    System.int OperationType,
    System.int ToolBodyCount,
    ref Body2 ToolBodyArr,
    System.int NumOfMatchingFaces,
    ref Face2 FaceList1,
    ref Face2 FaceList2,
    System.double MatchingTolerance,
    out System.int ErrorCode
)

```

C++/CLI

```

EnumBodies2^ IMatchedBoolean4(
    & System.int OperationType,
    & System.int ToolBodyCount,
    & Body2^% ToolBodyArr,
    & System.int NumOfMatchingFaces,
    & Face2^% FaceList1,
    & Face2^% FaceList2,
    & System.double MatchingTolerance,
    & [Out] System.int ErrorCode
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OperationType

One of the following operation types:

- SWBODYADD
- SWBODYCUT
- SWBODYINTERSECT

ToolBodyCount

Number of bodies

ToolBodyArr

Array of [bodies](#) of size toolBodyCount

NumOfMatchingFaces

Number of matching faces

FaceList1

First face list (see **Remarks**)

FaceList2

Second face list (see **Remarks**)

MatchingTolerance

Tolerance to use to check matching faces

ErrorCode

Error indicated as defined in [swBodyOperationError_e](#)

Visual Basic for Applications (VBA) Syntax

See [Body2::IMatchedBoolean4](#).

Remarks

The concept of match means that the caller tells the boolean operator beforehand which faces can be considered to coincide. Basically the caller performs part of the boolean operation.

Sometimes the application knows that two faces match because of the way the bodies are constructed; i.e., the application knows which faces are intended to match.

Having a list of matching face pairs may allow the matched boolean operator to resolve other geometric operations that otherwise it would not be able to work out. In general, providing matched faces speeds up the boolean operation and makes results more reliable.

The arguments FaceList1 and FaceList2 arguments can be empty lists. If matching face pairs are passed in, these faces must match such that:

- the surface geometry is coinciding.
- for each edge in a face, there is an edge in the other face that coincides.

If MatchingTolerance is less than 1.0e-8 or 0.0, then a default tolerance of 2.0e-6 is used. You decide the tolerance value based on the similarities and subtle differences between the two bodies.

This method supports multibody parts.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::MatchedBoolean4 Method](#)

Availability

SOLIDWORKS 2008 SP1, Revision Number 16.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

IMaterialPropertyValues Property (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IMaterialPropertyValues Property (IBody2)

Obsolete. Superseded by [IBody2::IMaterialPropertyValues2](#).

.NET Syntax

Visual Basic (Declaration)

```
Property IMaterialPropertyValues As System.Double
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As System.Double  
  
instance.IMaterialPropertyValues = value  
  
value = instance.IMaterialPropertyValues
```

C#

```
System.double IMaterialPropertyValues {get; set;}
```

C++/CLI

```
property System.double IMaterialPropertyValues {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Body2::IMaterialPropertyValues](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IMaterialPropertyValues2 Property (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IMaterialPropertyValues2 Property (IBody2)

Gets or sets the material properties for a body other than the base body in the active configuration.

.NET Syntax

Visual Basic (Declaration)

Property IMaterialPropertyValues2 As System.Double

Visual Basic (Usage)



```
Dim instance As IBody2
Dim value As System.Double

instance.IMaterialPropertyValues2 = value

value = instance.IMaterialPropertyValues2
```

C#

```
System.double IMaterialPropertyValues2 {get; set;}
```

C++/CLI

```
property System.double IMaterialPropertyValues2 {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

- in-process, unmanaged C++: Pointer to array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The material values include the face color (R,G,B values), reflectivity (ambient, diffuse, specular, shininess), transparency, and emission. Valid settings are from 0 to 1 for all values.

This property is intended to be used on bodies other than the base body and should follow a call to [IPartDoc::EnumRelatedBodies2](#). The format of the parameters or return values is an array of 9 doubles as follows:

[*R*, *G*, *B*, *Ambient*, *Diffuse*, *Specular*, *Shininess*, *Transparency*, *Emission*]

To see a color change, you must:

1. Specify *R*, *G*, and *B*, each with a value between 0.0 and 1.0, inclusive. (These values are internally multiplied by 255.0 to determine the RGB color.)
2. Specify the reflectivity properties (*Diffuse*, *Specular*, *Shininess* (1.0 - Specular spread/Blurriness)), each with a value greater than 0.0 and less than or equal to 1.0.
3. Specify *Ambient*, *Transparency* and *Emission*, each with a value between 0.0 and 1.0, inclusive.
4. Refresh the graphics area after setting this property.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::MaterialPropertyValues2 Property](#)
[IComponent2::IMaterialPropertyValues Property](#)
[IFace2::IMaterialPropertyValues Property](#)
[IModelDoc2::IMaterialPropertyValues Property](#)
[IPartDoc::IMaterialPropertyValues Property](#)
[IBody2::GetMaterialIdName2 Method](#)
[IBody2::GetMaterialPropertyName Method](#)
[IBody2::GetMaterialUserName2 Method](#)
[IBody2::IGetMaterialPropertyValuesForFace Method](#)
[IBody2::IRemoveMaterialProperty Method](#)
[IBody2::RemoveMaterialProperty Method](#)
[IBody2::SetMaterialIdName2 Method](#)
[IBody2::SetMaterialProperty Method](#)
[IBody2::SetMaterialUserName2 Method](#)

IOperations Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IOperations Method (IBody2)

Obsolete. Superseded by [IBody2::Operations2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IOperations( _  
    ByVal OperationType As System.Integer, _  
    ByVal ToolBody As Body2, _  
    ByVal NumMaxSections As System.Integer, _  
    ByRef ResultingBodies As Body2 _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim OperationType As System.Integer  
Dim ToolBody As Body2  
Dim NumMaxSections As System.Integer  
Dim ResultingBodies As Body2  
Dim value As System.Integer  
  
value = instance.IOperations(OperationType, ToolBody, NumMaxSections, ResultingBodies)
```

C#

```
System.int IOperations(  
    System.int OperationType,  
    Body2 ToolBody,  
    System.int NumMaxSections,  
    ref Body2 ResultingBodies  
)
```

C++/CLI

```
System.int IOperations(  
& System.int OperationType,  
& Body2^ ToolBody,  
& System.int NumMaxSections,  
& Body2^% ResultingBodies  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OperationType

ToolBody

NumMaxSections

ResultingBodies

Visual Basic for Applications (VBA) Syntax

See [Body2::IOperations](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)



SolidWorks.Interop.sldworks Namespace > [IBody2 Interface](#) : IOperations2 Method (IBody2)

Performs add, cut, and intersect (unite, subtract, and interfere) operations between two temporary bodies.

.NET Syntax

Visual Basic (Declaration)

```
Function IOperations2( _  
    ByVal OperationType As System.Integer, _  
    ByVal ToolBody As Body2, _  
    ByRef ErrorCode As System.Integer _  
) As EnumBodies2
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim OperationType As System.Integer  
Dim ToolBody As Body2  
Dim ErrorCode As System.Integer  
Dim value As EnumBodies2  
  
value = instance.IOperations2(OperationType, ToolBody, ErrorCode)
```

C#

```
EnumBodies2 IOperations2(  
    System.int OperationType,  
    Body2 ToolBody,  
    out System.int ErrorCode  
)
```

C++/CLI

```
EnumBodies2^ IOperations2(  
& System.int OperationType,  
& Body2^ ToolBody,  
& [Out] System.int ErrorCode  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OperationType

Operation type as defined in [swBodyOperationType_e](#)

ToolBody

Pointer to the tool [body](#)

ErrorCode

Error indicator as defined in [swBodyOperationError_e](#); returns swBodyOperationNoError if SOLIDWORKS does not generate an error

Return Value

Resulting [bodies enumeration](#)

Visual Basic for Applications (VBA) Syntax

See [Body2::IOperations2](#).

Remarks

If the target and tool bodies are the same in geometry, the result body of this method is NULL, and the return value is S_false.

This method works with two temporary bodies; one is the target and one is the tool. The output is a list of bodies resulting from the operation.

The two temporary bodies used in this function (the Body2 and ToolBody pointers) are invalid once the operation is complete. COM applications should release these two pointers after using this method. If your application needs to maintain these bodies, then you should make a copy of them using [IBody2::ICopy](#) before passing them to this routine.

To perform a [swBodyOperationType_e](#).SWBODYINTERSECT between a sheet (surface) and a solid body, the sheet body must be the target body.

Use [IBody2::Check3](#) for both bodies before using this method to ensure that both bodies are valid solids. Using this method with invalid bodies can cause unexpected results.

If a non-manifold error is returned, use [IBody2::ResetEdgeTolerances](#) to visit all of the edges in the body to reset their tolerances. Then use [IBody2::Operations2](#) again.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::Operations2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IRemoveFacesFromSheet Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IRemoveFacesFromSheet Method (IBody2)

Removes the specified faces from a sheet (surface) body.

.NET Syntax

Visual Basic (Declaration)

```
Sub IRemoveFacesFromSheet( _  
    ByVal NumOfFaces As System.Integer, _  
    ByRef FacesToRemove As Face2 _  
)
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim NumOfFaces As System.Integer  
Dim FacesToRemove As Face2
```

```
instance.IRemoveFacesFromSheet(NumOfFaces, FacesToRemove)
```

C#

```
void IRemoveFacesFromSheet(  
    System.int NumOfFaces,  
    ref Face2 FacesToRemove  
)
```

C++/CLI

```
void IRemoveFacesFromSheet(  
    & System.int NumOfFaces,  
    & Face2^% FacesToRemove  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumOfFaces

Number of faces generated when these two bodies intersect

FacesToRemove

Pointer to an array of [faces](#) to remove

Visual Basic for Applications (VBA) Syntax

See [Body2::IRemoveFacesFromSheet](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::RemoveFacesFromSheet Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IRemoveMaterialProperty Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IRemoveMaterialProperty Method (IBody2)

Removes the material property values (e.g., color) from the body in the specified configurations in parts and assemblies.

.NET Syntax

Visual Basic (Declaration)

```
Function IRemoveMaterialProperty( _  
    ByVal Config\_opt As System.Integer, _  
    ByVal Config\_count As System.Integer, _  
    ByRef Config\_names As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim Config_opt As System.Integer  
Dim Config_count As System.Integer  
Dim Config_names As System.String  
Dim value As System.Boolean  
  
value = instance.IRemoveMaterialProperty(Config_opt, Config_count, Config_names)
```

C#

```
System.bool IRemoveMaterialProperty(  
    System.int Config\_opt,  
    System.int Config\_count,  
    ref System.string Config\_names  
)
```

C++/CLI

```
System.bool IRemoveMaterialProperty(  
& System.int Config\_opt,  
& System.int Config\_count,  
& System.String^% Config\_names  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Config_opt
Configuration option as defined in [swInConfigurationOpts_e](#)
Config_count
Number of configurations in the part or assembly
Config_names
Array of configuration names

Return Value

True if the material is removed from the body, false if not

Visual Basic for Applications (VBA) Syntax

See [Body2::IRemoveMaterialProperty](#).

Remarks

Before calling this method, call [IModelDoc2::GetConfigurationCount](#) to get the value for Config_Count.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::RemoveMaterialProperty Method](#)
[IBody2::IGetMaterialPropertyValuesForFace Method](#)
[IBody2::IMaterialPropertyValues2 Property](#)
[IBody2::MaterialPropertyValues2 Property](#)
[IBody2::GetMaterialIdName2 Method](#)
[IBody2::GetMaterialPropertyName2 Method](#)
[IBody2::SetMaterialIdName2 Method](#)
[IBody2::SetMaterialPropertyName2 Method](#)
[IBody2::SetMaterialUserName2 Method](#)
[IBody2::GetMaterialUserName2 Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISave Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : ISave Method (IBody2)

Saves this body.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISave( _  
    ByVal StreamIn As System.Object _  
)
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim StreamIn As System.Object  
  
instance.ISave(StreamIn)
```

C#

```
void ISave(  
    System.object StreamIn  
)
```

C++/CLI

```
void ISave(  
& System.Object^ StreamIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

StreamIn

Stream to use for the save

Visual Basic for Applications (VBA) Syntax

See [Body2::ISave](#).

Remarks

If you want to save the solid body associated with the document, then save the document.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::Save Method](#)
[IModeler::IRestore2 Method\(\)](#)
[IModeler::Restore Method\(\)](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISectionBySheet Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : ISectionBySheet Method (IBody2)

Sections a body using a sheet (surface) body.

.NET Syntax

Visual Basic (Declaration)

```
Function ISectionBySheet( _  
    ByVal Sheet As Body2, _  
    ByVal NumMaxSections As System.Integer, _  
    ByRef SectionedBodies As Body2 _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim Sheet As Body2  
Dim NumMaxSections As System.Integer  
Dim SectionedBodies As Body2  
Dim value As System.Integer  
  
value = instance.ISectionBySheet(Sheet, NumMaxSections, SectionedBodies)
```

C#

```
System.int ISectionBySheet(  
    Body2 Sheet,  
    System.int NumMaxSections,  
    ref Body2 SectionedBodies  
)
```

C++/CLI

```
System.int ISectionBySheet(  
& Body2^ Sheet,  
& System.int NumMaxSections,  
& Body2^% SectionedBodies  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Sheet

Pointer to the sheet body used to perform the section

NumMaxSections

Maximum number of sections to create

SectionedBodies

Pointer to an array of [bodies](#) created during the section operation

Return Value

Number of bodies created during the operation

Visual Basic for Applications (VBA) Syntax

See [Body2::!SectionBySheet](#).

Remarks

Before using this method, [make a copy of the body](#) because the sheet body becomes invalid after using this method. COM applications should release all bodies after calling this method.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetCurrentSurface Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : ISetCurrentSurface Method (IBody2)

Places an existing surface into a temporary body that is under construction.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetCurrentSurface( _  
    ByVal SurfaceIn As Surface _  
)
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim SurfaceIn As Surface  
  
instance.ISetCurrentSurface(SurfaceIn)
```

C#

```
void ISetCurrentSurface(  
    Surface SurfaceIn  
)
```

C++/CLI

```
void ISetCurrentSurface(  
& Surface^ SurfaceIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SurfaceIn

Pointer to a [surface](#), which might have been created using other surface creation routines, such as [IModeler::!CreateCylindricalSurface2](#)

Visual Basic for Applications (VBA) Syntax

See [Body2::!SetCurrentSurface](#).

Remarks

This method is used with a set of related methods that construct a body from trimmed surfaces. This method takes an ISurface object created elsewhere and adds it to the temporary body object, which acts as a placeholder for the trimmed surfaces.

Follow calls to this method with one or more calls to the trimming-curve creation methods, such as [ISurface::AddTrimmingLoop2](#). Then, trim the surface using [IBody2::CreateTrimmedSurface](#). After you add all the surfaces to the body and trim appropriately, you can sew the body to create an imported SOLIDWORKS body feature using [IBody2::CreateBodyFromSurfaces](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::SetCurrentSurface Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetXform Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : ISetXform Method (IBody2)

Obsolete. Superseded by [IBody2::ApplyTransform](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ISetXform(  
    ByRef XformIn As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim XformIn As System.Double  
Dim value As System.Boolean  
  
value = instance.ISetXform(XformIn)
```

C#

```
System.bool ISetXform(  
    ref System.double XformIn  
)
```

C++/CLI

```
System.bool ISetXform(  
    & System.double% XformIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XformIn

Visual Basic for Applications (VBA) Syntax

See [Body2::ISetXform](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsGraphicsBody Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IsGraphicsBody Method (IBody2)

Gets whether this body is a graphics body.

.NET Syntax

Visual Basic (Declaration)

```
Function IsGraphicsBody() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As System.Boolean  
  
value = instance.IsGraphicsBody()
```

C#

```
System.bool IsGraphicsBody()
```

C++/CLI

```
System.bool IsGraphicsBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if a graphics body, false if not

Visual Basic for Applications (VBA) Syntax

See [Body2::IsGraphicsBody](#).

Example

See the [IFacet](#) examples.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetGraphicsBody Method \(\)](#)

Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsMeshBody Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IsMeshBody Method (IBody2)

Gets whether this body is a mesh body.

.NET Syntax

Visual Basic (Declaration)

```
Function IsMeshBody() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As System.Boolean
```

```
value = instance.IsMeshBody()
```

C#

```
System.bool IsMeshBody()
```

C++/CLI

```
System.bool IsMeshBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if a mesh body, false if not

Visual Basic for Applications (VBA) Syntax

See [Body2::IsMeshBody](#).

Example

See the [IFacet](#) examples.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetMeshBody Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsPatternSeed Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IsPatternSeed Method (IBody2)

Gets whether this body is the seed of a patterned body.

.NET Syntax

Visual Basic (Declaration)

```
Function IsPatternSeed( _  
    ByVal BodyDispIn As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim BodyDispIn As System.Object  
Dim value As System.Boolean
```

```
value = instance.IsPatternSeed(BodyDispIn)
```

C#

```
System.bool IsPatternSeed(  
    System.object BodyDispIn  
)
```

C++/CLI

```
System.bool IsPatternSeed(  
& System.Object^ BodyDispIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BodyDispIn
Patterned body

Return Value

True if this body is the seed of the patterned body, false if not

Visual Basic for Applications (VBA) Syntax

See [Body2::IsPatternSeed](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsSafe Property (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IsSafe Property (IBody2)

Not implemented.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property IsSafe As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As System.Boolean
```

```
value = instance.IsSafe
```

C#

```
System.bool IsSafe {get;}
```

C++/CLI

```
property System.bool IsSafe {  
    System.bool get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if the IBody2 object survived the rebuild, false if not

NOTE: This property is not implemented; thus, it will always return false.

Visual Basic for Applications (VBA) Syntax

See [Body2::IsSafe](#).

Remarks

This property is read-only and does not persist from session to session.

To make a body safe, use [IBody2::GetSafeBody](#) .

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsSheetMetal Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IsSheetMetal Method (IBody2)

Gets whether this body was created by a sheet metal feature.

.NET Syntax

Visual Basic (Declaration)

```
Function IsSheetMetal() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As System.Boolean  
  
value = instance.IsSheetMetal()
```

C#

```
System.bool IsSheetMetal()
```

C++/CLI

```
System.bool IsSheetMetal();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if this body was created by a sheet metal feature, false if not

Visual Basic for Applications (VBA) Syntax

See [Body2::IsSheetMetal](#).

Remarks

This method returns false for sheet metal bodies that are obtained from lightweight components using [IComponent2::GetBodies3](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2009 FCS , Revision Number 17.0

SOLIDWORKS API Help

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IsTemporaryBody Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : IsTemporaryBody Method (IBody2)

Gets whether the body is a temporary body.

.NET Syntax

Visual Basic (Declaration)

```
Function IsTemporaryBody() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim value As System.Boolean

value = instance.IsTemporaryBody()
```

C#

```
System.bool IsTemporaryBody()
```

C++/CLI

```
System.bool IsTemporaryBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True the body is a temporary body, false if not

Visual Basic for Applications (VBA) Syntax

See [Body2::IsTemporaryBody](#).

Example

[Delete Faces \(VBA\)](#)
[Create Loft Body \(C#\)](#)
[Create Loft Body \(VB.NET\)](#)
[Create Loft Body \(VBA\)](#)
[Delete Blended Faces \(C#\)](#)
[Delete Blended Faces \(VB.NET\)](#)
[Delete Blended Faces \(VBA\)](#)

Remarks

You can select an entity from a temporary body. You can also assign colors to faces on temporary bodies and to entire temporary bodies. See [IBody2::Display3](#) and [IFace2::MaterialPropertyValues](#) or [IFace2::IMaterialPropertyValues](#) and [IBody2::MaterialPropertyValues2](#) or [IBody2::IMaterialPropertyValues2](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12

SOLIDWORKS API Help

[Send comments on this topic.](#)

MakeOffset Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : MakeOffset Method (IBody2)

Creates a new temporary body by offsetting the selected surface body by the specified distance and in the specified direction.

.NET Syntax

Visual Basic (Declaration)

```
Function MakeOffset( _  
    ByVal Distance As System.Double, _  
    ByVal Direction As System.Boolean _  
) As Body2
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim Distance As System.Double  
Dim Direction As System.Boolean  
Dim value As Body2  
  
value = instance.MakeOffset(Distance, Direction)
```

C#

```
Body2 MakeOffset(  
    System.double Distance,  
    System.bool Direction  
)
```

C++/CLI

```
Body2^ MakeOffset(  
& System.double Distance,  
& System.bool Direction  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Distance

Distance by which to offset the selected surface body

Direction

True to offset the selected surface body in the opposite direction, false to offset the surface body along the normal

Return Value

Pointer to the newly created [body](#)

Visual Basic for Applications (VBA) Syntax

See [Body2::MakeOffset](#).

Example

[Create Temporary Bodies by Offsetting Surface Body \(C#\)](#)
[Create Temporary Bodies by Offsetting Surface Body \(VB.NET\)](#)
[Create Temporary Bodies By Offsetting a Surface Body \(VBA\)](#)

Remarks

This method only supports surface bodies.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14



.NET Syntax

Visual Basic (Declaration)

```
Function MatchedBoolean( _  
    ByVal OperationType As System.Integer, _  
    ByVal ToolBody As System.Object, _  
    ByVal NumOfMatchingFaces As System.Integer, _  
    ByVal FaceList1 As System.Object, _  
    ByVal FaceList2 As System.Object, _  
    ByRef ErrorCode As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim OperationType As System.Integer  
Dim ToolBody As System.Object  
Dim NumOfMatchingFaces As System.Integer  
Dim FaceList1 As System.Object  
Dim FaceList2 As System.Object  
Dim ErrorCode As System.Integer  
Dim value As System.Object  
  
value = instance.MatchedBoolean(OperationType, ToolBody, NumOfMatchingFaces, FaceList1, FaceList2, ErrorCode)
```

C#

```
System.object MatchedBoolean(  
    System.int OperationType,  
    System.object ToolBody,  
    System.int NumOfMatchingFaces,  
    System.object FaceList1,  
    System.object FaceList2,  
    out System.int ErrorCode  
)
```

C++/CLI

```
System.Object^ MatchedBoolean(  
& System.int OperationType,  
& System.Object^ ToolBody,  
& System.int NumOfMatchingFaces,  
& System.Object^ FaceList1,  
& System.Object^ FaceList2,  
& [Out] System.int ErrorCode  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OperationType

ToolBody

NumOfMatchingFaces

FaceList1

FaceList2

ErrorCode

Visual Basic for Applications (VBA) Syntax

See [Body2::MatchedBoolean](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

SOLIDWORKS API Help

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MatchedBoolean2 Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : MatchedBoolean2 Method (IBody2)

Obsolete. Superseded by [IBody2::MatchedBoolean3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function MatchedBoolean2(  
    ByVal OperationType As System.Integer, _  
    ByVal ToolBody As System.Object, _  
    ByVal NumOfMatchingFaces As System.Integer, _  
    ByVal FaceList1 As System.Object, _  
    ByVal FaceList2 As System.Object, _  
    ByRef ErrorCode As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim OperationType As System.Integer  
Dim ToolBody As System.Object  
Dim NumOfMatchingFaces As System.Integer  
Dim FaceList1 As System.Object  
Dim FaceList2 As System.Object  
Dim ErrorCode As System.Integer  
Dim value As System.Object  
  
value = instance.MatchedBoolean2(OperationType, ToolBody, NumOfMatchingFaces, FaceList1,  
FaceList2, ErrorCode)
```

C#

```
System.object MatchedBoolean2(  
    System.int OperationType,  
    System.object ToolBody,  
    System.int NumOfMatchingFaces,  
    System.object FaceList1,  
    System.object FaceList2,  
    out System.int ErrorCode  
)
```

C++/CLI

```
System.Object^ MatchedBoolean2(  
& System.int OperationType,  
& System.Object^ ToolBody,  
& System.int NumOfMatchingFaces,  
& System.Object^ FaceList1,  
& System.Object^ FaceList2,  
& [Out] System.int ErrorCode  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OperationType

ToolBody

NumOfMatchingFaces

FaceList1

FaceList2

ErrorCode

Visual Basic for Applications (VBA) Syntax

See [Body2::MatchedBoolean2](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

MatchedBoolean3 Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : MatchedBoolean3 Method (IBody2)

Obsolete. Superseded by [IBody2::MatchedBoolean4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function MatchedBoolean3( _  
    ByVal OperationType As System.Integer, _  
    ByVal ToolBody As System.Object, _  
    ByVal NumOfMatchingFaces As System.Integer, _  
    ByVal FaceList1 As System.Object, _  
    ByVal FaceList2 As System.Object, _  
    ByRef ErrorCode As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim OperationType As System.Integer  
Dim ToolBody As System.Object  
Dim NumOfMatchingFaces As System.Integer  
Dim FaceList1 As System.Object  
Dim FaceList2 As System.Object  
Dim ErrorCode As System.Integer  
Dim value As System.Object  
  
value = instance.MatchedBoolean3(OperationType, ToolBody, NumOfMatchingFaces, FaceList1,  
FaceList2, ErrorCode)
```

C#

```
System.object MatchedBoolean3(  
    System.int OperationType,  
    System.object ToolBody,  
    System.int NumOfMatchingFaces,  
    System.object FaceList1,  
    System.object FaceList2,  
    out System.int ErrorCode  
)
```

C++/CLI

```
System.Object^ MatchedBoolean3(  
& System.int OperationType,  
& System.Object^ ToolBody,  
& System.int NumOfMatchingFaces,  
& System.Object^ FaceList1,  
& System.Object^ FaceList2,
```

```
& [Out] System.int ErrorCode  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OperationType

One of the following operation types:

- SWBODYYADD
- SWBODYCUT
- SWBODYINTERSECT

ToolBody

Array of bodies

NumOfMatchingFaces

Number of matching faces

FaceList1

First face list (see **Remarks**)

FaceList2

Second face list (see **Remarks**)

ErrorCode

Error indicated as defined in [swBodyOperationError_e](#)

Return Value

Array of [bodies](#) formed by the operation

Visual Basic for Applications (VBA) Syntax

See [Body2::MatchedBoolean3](#).

Remarks

The concept of match means that the caller tells the boolean operator beforehand which faces can be considered to coincide. Basically, the caller performs part of the boolean operation.

Sometimes the application knows that two faces match because of the way the bodies are constructed; i.e., the application knows which faces are intended to match.

Having a list of matching face pairs may allow the matched boolean operator to resolve other geometric operations that otherwise it would not be able to work out. In general, providing matched faces speeds up the boolean operation and makes results more reliable.

The arguments FaceList1 and FaceList2 arguments can be empty lists. If matching face pairs are passed in, these faces must match such that:

- the surface geometry is coinciding.
- for each edge in a face, there is an edge in the other face that coincides

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::IMatchedBoolean3 Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

MatchedBoolean4 Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : MatchedBoolean4 Method (IBody2)

Performs a matched boolean on the specified bodies and supports an optional list of faces that match exactly.

.NET Syntax

Visual Basic (Declaration)

```

Function MatchedBoolean4(
    ByVal OperationType As System.Integer, _
    ByVal ToolBody As System.Object, _
    ByVal NumOfMatchingFaces As System.Integer, _
    ByVal FaceList1 As System.Object, _
    ByVal FaceList2 As System.Object, _
    ByVal MatchingTolerance As System.Double, _
    ByRef ErrorCode As System.Integer _
) As System.Object

```

Visual Basic (Usage)



```

Dim instance As IBody2
Dim OperationType As System.Integer
Dim ToolBody As System.Object
Dim NumOfMatchingFaces As System.Integer
Dim FaceList1 As System.Object
Dim FaceList2 As System.Object
Dim MatchingTolerance As System.Double
Dim ErrorCode As System.Integer
Dim value As System.Object

value = instance.MatchedBoolean4(OperationType, ToolBody, NumOfMatchingFaces, FaceList1,
FaceList2, MatchingTolerance, ErrorCode)

```

C#

```

System.object MatchedBoolean4(
    System.int OperationType,
    System.object ToolBody,
    System.int NumOfMatchingFaces,
    System.object FaceList1,
    System.object FaceList2,
    System.double MatchingTolerance,
    out System.int ErrorCode
)

```

C++/CLI

```

System.Object^ MatchedBoolean4(
    & System.int OperationType,
    & System.Object^ ToolBody,
    & System.int NumOfMatchingFaces,
    & System.Object^ FaceList1,
    & System.Object^ FaceList2,
    & System.double MatchingTolerance,
    & [Out] System.int ErrorCode
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OperationType

One of the following operation types:

- SWBODYADD
- SWBODYCUT
- SWBODYINTERSECT

ToolBody

Array of bodies

NumOfMatchingFaces

Number of matching faces

FaceList1

First face list (see **Remarks**)

FaceList2

Number of matching faces

MatchingTolerance

Tolerance to use to check matching faces

ErrorCode

Error indicated as defined in [swBodyOperationError_e](#)

Visual Basic for Applications (VBA) Syntax

See [Body2::MatchedBoolean4](#).

Remarks

The concept of match means that the caller tells the boolean operator beforehand which faces can be considered to coincide. Basically the caller performs part of the boolean operation.

Sometimes the application knows that two faces match because of the way the bodies are constructed; i.e., the application knows which faces are intended to match.

Having a list of matching face pairs may allow the matched boolean operator to resolve other geometric operations that otherwise it would not be able to work out. In general, providing matched faces speeds up the boolean operation and makes results more reliable.

The arguments FaceList1 and FaceList2 arguments can be empty lists. If matching face pairs are passed in, these faces must match such that:

- the surface geometry is coinciding.
- for each edge in a face, there is an edge in the other face that coincides.

If MatchingTolerance is less than 1.0e-8 or 0.0, then a default tolerance of 2.0e-6 is used. You decide the tolerance value based on the similarities and subtle differences between the two bodies.

This method supports multibody parts.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::IMatchedBoolean4 Method](#)

Availability

SOLIDWORKS 2008 SP1, Revision Number 16.1